

Table 1: Comparative Analysis of RAG Paradigms

Paradigm	Key Features	Strengths
Naïve RAG	Keyword-based retrieval (e.g., TF-IDF, BM25)	- Simple and easy to implement - Suitable for fact-based queries
Advanced RAG	- Dense retrieval models (e.g., DPR) - Neural ranking and re-ranking - Multi-hop retrieval	- High precision retrieval - Improved contextual relevance
Modular RAG	- Hybrid retrieval (sparse and dense) - Tool and API integration - Composable, domain-specific pipelines	- High flexibility and customization - Suitable for diverse applications - Scalable
Graph RAG	- Integration of graph-based structures - Multi-hop reasoning - Contextual enrichment via nodes	- Relational reasoning capabilities - Mitigates hallucinations - Ideal for structured data tasks
Agentic RAG	- Autonomous agents - Dynamic decision-making - Iterative refinement and workflow optimization	- Adaptable to real-time changes - Scalable for multi-domain tasks - High accuracy

2.4.2 Multi-Step Reasoning

Many real-world queries require iterative or multi-hop reasoning—retrieving and synthesizing information across multiple steps. Traditional RAG systems are often ill-equipped to refine retrieval based on intermediate insights or user feedback, resulting in incomplete or disjointed responses.

Example: A complex query like, *"What lessons from renewable energy policies in Europe can be applied to developing nations, and what are the potential economic impacts?"* demands the orchestration of multiple types of information, including policy data, contextualization for developing regions, and economic analysis. Traditional RAG systems typically fail to connect these disparate elements into a cohesive response.

2.4.3 Scalability and Latency Issues

As the volume of external data sources grows, querying and ranking large datasets becomes increasingly computationally intensive. This results in significant latency, which undermines the system's ability to provide timely responses in real-time applications.

Example: In time-sensitive settings such as *financial analytics* or *live customer support*, delays caused by querying multiple databases or processing large document sets can hinder the system's overall utility. For example, a delay in retrieving market trends during high-frequency trading could result in missed opportunities.

2.5 Agentic RAG: A Paradigm Shift

Traditional RAG systems, with their static workflows and limited adaptability, often struggle to handle dynamic, multi-step reasoning and complex real-world tasks. These limitations have spurred the integration of agentic intelligence, resulting in Agentic RAG. By incorporating autonomous agents capable of dynamic decision-making, iterative reasoning, and adaptive retrieval strategies, Agentic RAG builds on the modularity of earlier paradigms while overcoming their inherent constraints. This evolution enables more complex, multi-domain tasks to be addressed with enhanced precision and contextual understanding, positioning Agentic RAG as a cornerstone for next-generation AI applications. In